

ENGR010 - S23 - EN 1-3

February 5, 2023

#

Weekly Engineering Assignment 1 - Group 3 - S23

A Demonstration Driveway Site

ENGR010SP23EN01GR03

** Color Code (nonfunctional in Google Colab): - External File/Imported Module - User-created Function - Python/Imported Function - Name-specific Data Structure/Variable

** Note: If you make your own markup cells instead of editing cells that start with “Edit this cell”, make sure no cell that starts with “Edit this cell” is left in the final version. Furthermore, if you edited the cells start with “Edit this cell”, make sure that the instructions from the original file are overwritten. Your submission should not contain any instructions.

0.1 Section 1

** You have just been hired as a summer intern at a engineering construction company. One of the company’s potential clients is a national home construction company. Your company would like to win a contract to build driveways for all the homebuilder’s new homes across the country. Your company is committed to the environment, and you are trying convince the client that concrete driveways are not only more durable than asphalt, but also more environmentally friendly. As part of the sales pitch, you have been asked to develop a proposal for a demonstration concrete driveway that can be shown to the client. The driveway will be 30 feet in length and should be the typical width of a two-car driveway (assume 24 feet). The concrete slab should be standard thickness for a driveway, which is 4 inches. Your boss has asked you to use the standard basic ingredients for the concrete rather than a premade mix.

** You want to be certain to do a good job on your very first project, so you write a paragraph that summarizes your understanding of the assignment, and then a second paragraph that outlines the steps you will take to complete it.

Edit this cell with a paragraph describing what the project is and what you have been asked to do. Remember, you can double click on any cell to edit it.

The boss asked me to create a proposal for a demonstration concrete driveway to win a contract for building driveways. At present, the specifications for the driveway are as follows: 1. The driveway is 30 feet in length. 2. It is 24 feet in width. 3. The thickness is 3 inches. 4. Standard basic ingredients for the concrete will be used, rather than a pre-made mix.

Edit this cell with a paragraph describing what steps you will need to take to complete the project.

The first step is to gather information on the ingredients to be used in the concrete. Next, estimate the amount of materials required based on the specifications. Then, determine the precise cost of the ingredients for the demonstration driveway model. Finally, the calculated cost is reviewed and approved by the boss and the construction of the demonstration driveway can commence.

****** You realize that before you do any calculations, you need to learn a little bit about concrete. Is it the same as cement? What is it even made of? Off you go to the internet, where you begin your learning. In the markdown cell below, summarize what you learned about concrete. Be sure to indicate your sources - number each source and put it in brackets; deally, you should create these as links so that someone just needs to click on the number to go directly to the linked site, like [1]. At the end of the report (not the end of the cell - the end of the assignment), list all the sources by number and URL so they can be found even if someone prints the report.

Edit this cell and type what you learned about concrete, including why it is an environmentally superior choice over asphalt.

I got the information from the following sites. [1] concretesupplyco.com [2] concretenetwork.com

A brief list of applicable information is as follows. 1. Composition: Concrete is made of cement, water, sand, and aggregates, while asphalt is a mixture of petroleum-based asphalt cement and aggregates. 2. Durability: Concrete is generally more durable and long-lasting than asphalt, but it can crack over time and require repairs. Asphalt is more flexible and less likely to crack, but it can develop potholes and wear down more quickly. 3. Maintenance: Asphalt typically requires more frequent sealing and maintenance, while concrete can be more low-maintenance. 4. Installation time: Concrete takes longer to install and cure than asphalt, but the longer curing time contributes to its greater durability. 5. Cost: Asphalt is generally less expensive to install than concrete, but the long-term costs of maintenance and repairs can be higher for asphalt than for concrete.

And I also looked up the video. [4] How to make concrete.

****** Now that you have learned a little about concrete and what it is, you need to calculate how much concrete you will need. Assume your driveway will be 4" thick, 30' long, and 24' wide. Use the cell below to calculate how much concrete you will need, in cubic feet. Be sure to use variable names that tell you what the variable is when you read the name.

```
[142]: thickness_inch = 4 #this value could be chaged to feet
length_feet = 30 #It can be modified according to your request.
width_feet = 24

inches_to_feet = 1/12

thickness_feet = thickness_inch * inches_to_feet

volume = thickness_feet * length_feet * width_feet

print("The amount of the concrete : %.2f cubic feet" %volume)
```

The amount of the concrete : 240.00 cubic feet

****** Your boss has asked you to make the concrete from its basic ingredients, so you need to estimate how much of each of these ingredients you will need to make the concrete. According to the cement

manufacturer, a 94 pound sack of Portland cement is 1.0 cubic feet in volume (dry volume, not to be confused with coverage volume) and makes 4.5 cubic feet of concrete using the standard 1:2:3 volumetric ratio of dry cement:sand:gravel. Use the cell below and any additional cells you would like to create to calculate the amount of each ingredient. To be prudent, add 10% to account for some potential waste, spillage etc.

```
[143]: # Calculate the volumes of ingredients to make 1 cubic feet of concrete.
# These values will be standard volume.

stand_cement = 1 / 4.5 # It means that to make 1 cubic feet of concrete, the
    ↪ cement will be used for 1/4.5 cubic feet.
stand_sand = stand_cement * 2 # Using 1:2:3 volumetric ratio
stand_gravel = stand_cement * 3 # Using 1:2:3 volumetric ratio
potential_waste = 1.1

# We need 240 cubic feet of concrete, also consider the potential waste
total_cement = stand_cement * volume * potential_waste
total_sand = stand_sand * volume * potential_waste
total_gravel = stand_gravel * volume * potential_waste

print("The capacity of the materials required is as follows.")
print("The amount of the cement: %.2f cubic feet" % total_cement)
print("The amount of the sand: %.2f cubic feet" % total_sand)
print("The amount of the gravel: %.2f cubic feet" % total_gravel)

print("-----")
# If we know the density of other ingredients(sand, gravel), can calculate the
    ↪ weight of ingredients.
print("The weight of the cement: %.2f pounds" % (total_cement * 94))

# Suppose the density and calculate the weight of sand and gravel.
density_gravel = 105 # in lb/ft3
density_sand = 100 # in lb/ft3

print("The weight of the gravel: %.2f pounds" % (total_gravel * density_gravel))
print("The weight of the sand: %.2f pounds" % (total_sand * density_sand))
```

The capacity of the materials required is as follows.

The amount of the cement: 58.67 cubic feet

The amount of the sand: 117.33 cubic feet

The amount of the gravel: 176.00 cubic feet

The weight of the cement: 5514.67 pounds

The weight of the gravel: 18480.00 pounds

The weight of the sand: 11733.33 pounds

** Of course, no engineering estimate is complete without the cost. Go back to the internet and find out how much these ingredients will cost. Since you will purchase these materials by the bag,

you'll have a partial bag of each material left over in just about every case. You can neglect the cost of the water, which is negligible in this case. Use the cell below to calculate the material cost for the proposed project.

```
[144]: # The weight of the materials required is as follows:
weight_cement_pound = total_cement * 94
weight_sand_pound = total_sand * density_sand
weight_gravel_pound = total_gravel * density_gravel

# According to Homedepot, the prices are as follows:
price_cement_bag = 18.67 #in dollars
price_sand_bag = 6.47 #in dollars
price_gravel_bag = 5.37 #in dollars

# Caculate How many bags do we need. Throw away the decimal point and do +1
total_cement_bags = int(weight_cement_pound / 94 + 1)
total_sand_bags = int(weight_sand_pound / 50 + 1)
total_gravel_bags = int(weight_gravel_pound / 50 + 1)

print("The total bags of cements: %d bags" % total_cement_bags)
print("The total bags of gravel: %d bags" % total_sand_bags)
print("The total bags of sand: %d bags" % total_gravel_bags)

# The total price will be
total_price = total_cement_bags * price_cement_bag + total_sand_bags * \
    price_sand_bag + total_gravel_bags * price_gravel_bag
print("The total price is $%.2f" %total_price)
```

```
The total bags of cements: 59 bags
The total bags of gravel: 235 bags
The total bags of sand: 370 bags
The total price is $4608.88
```

** Once you finished your calculations, you excitedly tell your boss that you have finished. He scans your figures, and then he says that now that he has thought about it some more, it might make sense to use a premixed concrete after all. He asks you go back and find out how much it would cost to do this instead of using the basic materials. Note: an 80 pound bag of premixed concrete (after mixing with water) fills a 2 square foot by 4" slab. In the cell below, calculate how many bags of premixed concrete you will need and how much it will cost to do the job.

```
[145]: total_concrete_premixed_pound = width_feet * length_feet * 80 / 2
total_concrete_premixed_bags = int(total_concrete_premixed_pound / 80 + 1)

price_concrete_bag = 5.03
total_price_premixed = price_concrete_bag * total_concrete_premixed_bags

print("When you use the premixed concrete, the price is as follow:")
```

```
print("The total_price is $%.2f" %total_price_premixed)
```

When you use the premixed concrete, the price is as follow:

The total_price is \$1815.83

** Your boss tells you that his labor cost is about \$100/hour, accounting for pay, benefits, taxes, etc. He realizes it might save some time to use the premixed concrete and asks you to recommend whether the project should use the basic or the premixed materials. Summarize your recommendation below.

Edit this cell with your recommendation on whether to use the basic materials or the premixed. Be sure to explain your reasons.

Ironically, using premixed concrete comes out to be much cheaper. This is because we have purchased all the materials from Home Depot and purchasing a large number of bags is very inefficient in terms of price. I searched for both the price of gravel and sand for concrete through Google search, which is not constant, but it was much cheaper than the price of Home Depot for bulk purchases. According to my calculation, it costs about \$1200 to buy and prepare all the materials(sand, gravel, cement) separately. In fact, in this construction, no one will buy the material from the Home Depot. So, I think the price of the material is 1200 dollars and I'm going to write my thoughts.

We have only looked at the price side of the material in this project, except for too many conditions. It requires equipment to make concrete and a technician, and numerous conditions are hidden, including taxes and transportation costs. Especially in this case, I think it is better to use premixed concrete because it is still building a small model (compared to large-scale construction) to win the contract. If the company can mix and store and transport concrete right away, it can be cheaper to make.

0.2 Section 2

After you tell the boss how much it will cost to build the driveway he specified, he tells you that actually, he has a budget of \$1500 and is willing to spend it all to build the longest 24-foot wide, 4 inch deep driveway that he can using the three raw materials He also found out that he already has 125 bags of gravel and 75 bags of sand left over from previous jobs, so he can use those instead of buying additional bags. He asks you how long a driveway he can build using raw materials, and how many bags of each material he will need to buy in addition to what he already has. Be sure to print appropriate units (i.e. length of driveway in feet, bags of gravel, bags of sand, etc.)

Hint: One simple way to approach this problem is to “sell the existing materials back” and add this money to your budget.

Edit this cell with your description of your code. Then type your code into the code cell below. Resale as much as the amount of ingredients already present. Add the money to the budget, and calculate the cost of building a one foot long road. It can be obtained by using the total cost of building a 30-foot road.

```
[146]: sell_gravel = 125 * 5.37
sell_sand = 75 * 6.47
total_budget = sell_sand + sell_gravel + 1500
print("Total sales amount = $%.2f" %(sell_gravel + sell_sand))
```

```

print("The budget will be : $%.2f " %(total_budget))

feet_price = total_price / 30
print("The money needed to build a foot long driveway: $%.2f" %feet_price)
print("Maximum length of road that can be made on current budget: %0.2f\
↪feet\n" %(total_budget / feet_price))

print("The number of bags required for sand gravel cement is as follows:")

#You only need to return to the first cell and enter 17.29 in length and click
↪the run all button.
print("The total bags of cements: 34 bags")
print("The total bags of gravel: 136 bags")
print("The total bags of sand: 214 bags")
print("Total Price: $2663.88")

print("There's a $7 error here, but it's not a significant number given that\
↪potential waste and road length are built approximately at 17 instead of\
↪exactly 17.29.")

```

Total sales amount = \$1156.50

The budget will be : \$2656.50

The money needed to build a foot long driveway: \$153.63

Maximum length of road that can be made on current budget: 17.29 feet

The number of bags required for sand gravel cement is as follows:

The total bags of cements: 34 bags

The total bags of gravel: 136 bags

The total bags of sand: 214 bags

Total Price: \$2663.88

There's a \$7 error here, but it's not a significant number given that potential waste and road length are built approximately at 17 instead of exactly 17.29.

0.3 Section 3

Write a function that returns the concrete volume for any length and width - pass these as parameters to the function. Write a second function that returns the total amounts of cement, sand and gravel required for a given volume of concrete using the standard 1:2:3 ratio. The input parameter to this function should be the volume of concrete, and the function should return the volumes of cement, sand and gravel required as either a tuple or a list. Finally, write a function to return the cost of a driveway when passed as parameters the required volumes of cement, sand and gravel. This function should use the prices for these materials that you found above in the previous sections. Now test your functions by writing a loop that prints the cost of 24-foot wide 4 inch deep driveways for lengths from 20 ft. to 36 ft., at increments of two feet.

Edit this cell with your description of your code. Then type your code into the code cell below. Actually, after solving section 1 and 2, all of this was completed. I'll organize everything to print this out nicely. I will also make it simpler using functions.

If subdivide the price_materials function, can also return the number of bags of necessary substances and the price for each material.

[147]: *#The numbers below are the given values. It can be treated as a constant.*

```
stand_cement = 1 / 4.5
stand_sand = stand_cement * 2
stand_gravel = stand_cement * 3
potential_waste = 1.1
density_gravel = 105 # in lb/ft3
density_sand = 100 # in lb/ft3
price_cement_bag = 18.67 #in dollars
price_sand_bag = 6.47 #in dollars
price_gravel_bag = 5.37 #in dollars
```

[148]: *# <function> inch to feet (for thickness)*

```
def Converter_inch_feet(a):
    a = a * (1/12)
    return a

# <function> returns the concrete volume for any length and width
def get_concrete_volume(a, b, c): #30, 24, 4
    c = Converter_inch_feet(c)
    volume = a * b * c
    return volume

# <function> returns the materials volume for volume of concrete
def get_materials_volume(a): # a is volume of concrete
    total_cement = stand_cement * a * potential_waste
    total_sand = stand_sand * a * potential_waste
    total_gravel = stand_gravel * a * potential_waste

    return [total_cement, total_sand, total_gravel]

def price_materials(lst_a):
    weight_cement_pound = lst_a[0] * 94
    weight_sand_pound = lst_a[1] * density_sand
    weight_gravel_pound = lst_a[2] * density_gravel

    total_cement_bags = int(weight_cement_pound / 94 + 1)
    total_sand_bags = int(weight_sand_pound / 50 + 1)
    total_gravel_bags = int(weight_gravel_pound / 50 + 1)

    total_price = total_cement_bags * price_cement_bag + total_sand_bags *
    ↪price_sand_bag + total_gravel_bags * price_gravel_bag

    return total_price
```

```
[151]: print("Information about the materials needed to build a road with x length.(It
        is fixed at 4 inches thick and 24 inches wide.)")
print("-----")
for i in range (20, 37, 2):
    concrete_volume = get_concrete_volume(i, 24, 4)
    lst_materials_volume = get_materials_volume(concrete_volume)
    total_price = price_materials(lst_materials_volume)

    print("Information when the length of the road is %d feets" %i)
    print("Volume of concrete required %.1f cubic feet:" %concrete_volume)
    print("The amount of the cement: %.2f cubic feet" % lst_materials_volume[0])
    print("The amount of the sand: %.2f cubic feet" % lst_materials_volume[1])
    print("The amount of the gravel: %.2f cubic feet" % lst_materials_volume[2])

    print("The total price: $%.2f" %total_price)
    print("-----")
```

Information about the materials needed to build a road with x length.(It is fixed at 4 inches thick and 24 inches wide.)

Information when the length of the road is 20 feets
Volume of concrete required 160.0 cubic feet:
The amount of the cement: 39.11 cubic feet
The amount of the sand: 78.22 cubic feet
The amount of the gravel: 117.33 cubic feet
The total price: \$3088.98

Information when the length of the road is 22 feets
Volume of concrete required 176.0 cubic feet:
The amount of the cement: 43.02 cubic feet
The amount of the sand: 86.04 cubic feet
The amount of the gravel: 129.07 cubic feet
The total price: \$3401.43

Information when the length of the road is 24 feets
Volume of concrete required 192.0 cubic feet:
The amount of the cement: 46.93 cubic feet
The amount of the sand: 93.87 cubic feet
The amount of the gravel: 140.80 cubic feet
The total price: \$3683.37

Information when the length of the road is 26 feets
Volume of concrete required 208.0 cubic feet:
The amount of the cement: 50.84 cubic feet
The amount of the sand: 101.69 cubic feet
The amount of the gravel: 152.53 cubic feet
The total price: \$3995.82

Information when the length of the road is 28 feet
Volume of concrete required 224.0 cubic feet:
The amount of the cement: 54.76 cubic feet
The amount of the sand: 109.51 cubic feet
The amount of the gravel: 164.27 cubic feet
The total price: \$4302.90

Information when the length of the road is 30 feet
Volume of concrete required 240.0 cubic feet:
The amount of the cement: 58.67 cubic feet
The amount of the sand: 117.33 cubic feet
The amount of the gravel: 176.00 cubic feet
The total price: \$4608.88

Information when the length of the road is 32 feet
Volume of concrete required 256.0 cubic feet:
The amount of the cement: 62.58 cubic feet
The amount of the sand: 125.16 cubic feet
The amount of the gravel: 187.73 cubic feet
The total price: \$4921.33

Information when the length of the road is 34 feet
Volume of concrete required 272.0 cubic feet:
The amount of the cement: 66.49 cubic feet
The amount of the sand: 132.98 cubic feet
The amount of the gravel: 199.47 cubic feet
The total price: \$5221.94

Information when the length of the road is 36 feet
Volume of concrete required 288.0 cubic feet:
The amount of the cement: 70.40 cubic feet
The amount of the sand: 140.80 cubic feet
The amount of the gravel: 211.20 cubic feet
The total price: \$5534.39

** In the cell below, include a list of all the sources you used for this proposal. Be sure to indicate the date you used the site.

Edit this cell with a list of the sources you used for this assignment. I got the information on the price of the materials needed to make concrete through the link you gave below. The only content what I searched separately are the density of gravel and sand. 1. <https://civiltoday.com/civil-engineering-materials/sand/299-bulk-density-of-sand> 2. <https://www.gigacalculator.com/calculators/gravel-calculator.php>

** Here are some helpful sources to get your started. You do not have to use these!

1. <https://concretesupplyco.com/concrete-vs-asphalt/>
2. <https://www.cement.org/cement-concrete/paving/buildings-structures/concrete->

homes/products/driveways

3. <https://www.homedepot.com/c/ab/types-of-concrete-mix-for-any-project/9ba683603be9fa5395fab901c07575fe>
4. <https://images.homedepot-static.com/catalog/pdfImages/2c/2cd086ef-e650-44c8-a0d1-f5fc3ce4418b.pdf>
5. <https://www.homedepot.com/p/Quikrete-50-lb-All-Purpose-Sand-115251/100318450>
6. https://www.homedepot.com/p/Quikrete-50-lb-All-Purpose-Gravel-115150/100318444?source=shoppingads&locale=en-US&mtc=Shopping-B-F_D22-G-D22-22_9_CONCRETE-Generic-NA-Feed-LIA-NA-NA-&cm_mmc=Shopping-B-F_D22-G-D22-22_9_CONCRETE-Generic-NA-Feed-LIA-NA-NA-71700000045600364-58700004679444416-92700057047190399&gclid=CjwKCAjwIYCHBhAQEiwA4K21mx5M8-GpeTS8ZVBc4fBi23-RYiAK9x63VNeutjWOj6N8EctAaIsQORoCTh4QAvD_BwE&gclsrc=aw.ds
7. https://www.homedepot.com/p/94-lb-Portland-Cement-112494/100570364?source=shoppingads&locale=en-US&mtc=Shopping-B-F_D22-G-D22-22_9_CONCRETE-Generic-NA-Feed-LIA-NA-NA-&cm_mmc=Shopping-B-F_D22-G-D22-22_9_CONCRETE-Generic-NA-Feed-LIA-NA-NA-71700000045600364-58700004679444416-92700057047190396&gclid=CjwKCAjwIYCHBhAQEiwA4K21m7jd2TuCQAsmI-YdxhQrPKS0jL0Ai-V-R5GAHtcmBd0OdX5W9opHARoCkk8QAvD_BwE&gclsrc=aw.ds

** You are just about to submit your proposal. Go back through it and delete all the cells with instructions (they all begin with two asterisks). Add any additional formatting needed to make your proposal look professional. When you are finished, there should not be any cells that say “Edit this cell” because either you have edited those cells and replaced that text or you have deleted them. Double check all your assumptions and calculations to make sure your results are visible, reasonable and correct. If you can’t see your results, we won’t either, and this will hurt your grade. Next, do a print preview and make sure everything looks as you expected. Finally, save the print preview as a PDF (.pdf) file and submit both the both the JupyterNotebook (.ipynb) and PDF (.pdf) files to CourseSite. Congratulations on completing your Engineering Assignment 1.

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